TrustLens AI — Judge Quick Guide

Team Reshiram · DEX26223 · Open Innovation (Software) · DEXATHON 2026

Live: <link href="https://trustlensai-u2r6.vercel.app" color="#0E6FBF">trustlensai-u2r6.vercel.app</link> · Repo: <link href="https://github.com/trustlensaiofficial/TrustlensAi" color="#0E6FBF">github.com/trustlensaiofficial/TrustlensAi</link>

> The one-line story. Most tools show a red warning and stop. TrustLens shows you the exact words that are wrong, what it could and could not check, and one safe thing to do next — before you pay, click or share.

Tagline: Pause. Verify. Then act.

1 · Technology actually in the repository

Layer	Technology	Purpose
Frontend	Next.js 16.4 App Router, React 19.3, TypeScript 5.9	All pages and UI
Backend	Next.js Route Handlers (Node runtime)	/api/analyze, /api/health, /api/assistant, /api/report
Detection	TypeScript, no ML	43 fixed rules (17 generic + 19 context + 7 Tamil)
Database	Supabase PostgreSQL	Saved reports, Row Level Security
On device	IndexedDB	History that works offline, with no account
AI model	Claude Sonnet 5 via Vercel AI Gateway (AI SDK 6)	Plain-language explanation layer only
Validation	Zod 4	Every API input schema-validated
Auth	Supabase Auth — Google OAuth + passkeys	Optional sign-in; no passwords
Hosting	Vercel	Production deployment
Tests	Vitest — 185 passing	Detection, privacy, campaign logic

> ■ The one PPT mismatch — say this first, before a judge finds it.

“Our slide says Python Flask back end. That was the proposal-stage plan. In the code we shipped there is no Flask and no Python — the backend is Next.js Route Handlers on Node. We consolidated to one TypeScript codebase so the detection rules are shared by frontend and backend and covered by one test suite. We would rather correct the slide than claim a technology we did not use.”

2 · The flow

1 Input	Paste a message, link or claim. Pick what it is — bank/UPI, chat, link, job offer.
2 Warning signs	43 fixed rules scan the wording; each match records the EXACT phrase. Urgency, impersonation, payment demands, credential requests, risky links.
3 Evidence check	Link structure offline. Optionally DNS + page metadata over the network, with SSRF protection — private/loopback refused, every redirect re-validated, timeouts and size caps.
4 Report	Findings split by HOW each was established and never blended: Fixed rule · Checked · AI reading. Plus what could NOT be checked.
5 Verify first	One concrete step — usually “call your bank on the number printed on your card”. If money moved → 1930 first.

3 · Fifteen judge questions

1. What is TrustLens AI?

A personal financial-threat protection tool. Paste a suspicious message and it shows the exact warning signs, what it could and could not verify, and the safest next step.

2. What problem does it solve?

■22,845 crore lost to cyber fraud in India in 2024 — a 206% rise. Existing tools warn without evidence, so people still don't know what to do.

3. What is innovative?

Three things. Every finding quotes the exact triggering words. Findings are separated by how they were established — fixed rule, real check, or AI guess — and never merged. And we say what we could not check, instead of letting silence imply safety.

4. How is it different from Truecaller?

Truecaller tells you who is calling. We tell you whether the message deserves your trust and what to do — with the evidence shown. A different problem.

5. Which languages and frameworks?

TypeScript end to end. Next.js 16 App Router for frontend and backend, React 19, Supabase PostgreSQL, Claude Sonnet 5 via Vercel AI Gateway, deployed on Vercel.

6. How does scam detection work?

43 deterministic rules in three sets — 17 generic, 19 context-specific (bank, invoice, job, investment), 7 Tamil. Each records the exact matched span, which the report shows.

7. How do AI and rules work together?

Rules run first and always. The AI only adds a plain-language reading on top, labelled separately. If the AI is unavailable the product still works — deliberately.

8. How do you verify evidence?

Offline link-structure analysis always. Optionally live DNS and page metadata, with SSRF guards. We never claim a check ran when it did not.

9. How do you avoid false positives?

Benign controls are in the test suite and must stay silent — e.g. "I got the OTP for the train booking". We found and fixed exactly that: a critical rule matched the bare word "OTP". Now every credential rule requires a demand, not a mention.

10. How do you protect privacy?

Reports are stored on your own device in IndexedDB and work with no account. Nothing reaches the campaign network unless you choose. We do not claim end-to-end encryption — the server reads the text to analyse it, and our Privacy page says so.

11. What if AI is unavailable?

Everything except the AI explanation. Right now the gateway refuses on billing and the app says so on screen rather than showing a green tick. Rules, evidence, next steps and reports all work. You can see it live.

12. Current limitations?

No Android app — no SDK on our machine, not attempted rather than faked. Campaign data is simulated and labelled on every card. AI layer blocked on billing. We also cannot establish a sender's identity or location, and the report says so.

13. How can this scale?

The engine is pure functions with no per-user state, so it scales horizontally on serverless. Campaign matching is union-find over normalised indicators — near-linear. Phone numbers share as HMAC-keyed tokens, never raw or plain-hashed.

14. Business model?

Free core checks for students, individuals and families. Small-business plan for team accounts and payment-request checks. Partnerships with colleges and community groups. Costs are AI usage, lookups and hosting — no revenue figures assumed.

15. Why select you?

Because we built the honest version. It is easy to demo something that always says "SCAM DETECTED". We built one that tells you what it could not check, labels simulated data as simulated, and shows "AI unavailable" on stage rather than faking it. That restraint is the hard part — and the only version anyone could trust with their money.

4 · The 90-second explanation (all five members)

Last year Indians lost ■22,845 crore to cyber fraud — up 206%.

The tools that exist show a red warning and stop there. That creates panic without understanding, so people still don't know whether to pay, or what to do if they already did.

TrustLens does three things differently. One: every warning quotes the exact words that triggered it. Two: we separate a fixed rule from a real check from an AI guess — and never blend them. Three: we tell you what we could not check, because no warning found is not the same as safe.

It works in English, Tamil and Hindi — because a scam in India arrives as "OTP share pannunga", and English-only rules miss it entirely.

If something already happened it switches to incident response — and if money has moved it leads with 1930, because the first hours decide whether a transfer can be frozen.

It's live, works without an account, and the rules keep working with no internet.

Pause. Verify. Then act.

5 · Three-minute live demo

#	Click	Say
1	Home	"This is live. The status strip says AI layer unavailable — we'd rather show that than fake a green tick."
2	How it works → 2 scenes → Try it now	"A narrated walkthrough of every capability."
3	Check → example "Bank phishing SMS with a lookalike link"	"A real Indian bank scam, same pipeline as your own text."
4	The verdict	"Strong warning signs — and it tells you why."
5	Expand a warning sign	"It quotes the exact words. And here — hdfcbank is in the address, but the real domain is kyc-update-secure.xyz."
6	"How this was established"	"Three columns — fixed rule, actually checked, AI reading. Never blended. That's our core idea."
7	"What we could not check"	"The part most tools skip. No warning found is not the same as safe."
8	Next steps	"One concrete action: call your bank on the number printed on your card."
9	Clear → paste benign → Check	"An ordinary message mentioning an OTP. No signals detected. A tool that cries wolf gets ignored."
10	Respond → "I transferred money"	"1930 comes first, above everything — the first hours decide whether a transfer can be frozen."
11	Report → Prepare an evidence summary	"A summary for your bank or a complaint. You choose what it includes — and it says plainly it is not an official police document."
12	Scam alerts	"Campaign matching. Clearly labelled simulated — but the clustering is genuinely computed."

Benign message (step 9): I got the OTP for the train booking, the ticket is confirmed. See you Sunday.

Scam message: Dear customer, your HDFC account will be blocked today. Complete KYC immediately at http://kyc-update-secure.xyz and share the OTP with our executive.

6 · PPT vs live prototype — every honest gap

PPT says	Reality	Tell judges
Python Flask back end	No Flask, no Python. Next.js Route Handlers on Node	"Proposal-stage plan. We consolidated to one TypeScript codebase so rules are shared and tested once."

PPT says	Reality	Tell judges
Official-source and fact-check lookups	Link/DNS/metadata checks ARE live. Named fact-check APIs are NOT integrated	“Link and domain checks are live. Fact-check API integration is proposed, not built — and we never claim a source was consulted when it wasn’t.”
Source citations	Evidence rows show source, method, timestamp, status	“Implemented for the checks we actually run.”
AI explanations	Built and wired; gateway refuses on billing	“The layer is built. Our gateway needs billing enabled. The app shows AI layer unavailable rather than pretending.”
Community campaign alerts	Engine real and tested; data synthetic	“Matching, de-duplication and corroboration are genuinely computed. Reports are simulated and every card says so.”
—	No Android app	“Not attempted — no Android SDK on our build machine. We won’t show a simulator and call it a device integration.”

7 · Numbers you can state safely

- 43 deterministic rules (17 generic + 19 context + 7 Tamil)
- 185 automated tests passing
- 3 explanation languages: English, Tamil, Hindi
- ■22,845 crore lost to cyber fraud in India, 2024 (+206%) — MHA
- US$15.9B reported US fraud losses, 2025 — FTC
- 1930 — India’s cyber-financial fraud helpline

> Never state a detection accuracy percentage, user numbers, or that any message is “100% safe”. We have not measured accuracy on a labelled corpus — and saying so is the honest answer.

8 · If something fails on stage

- AI / Kaval errors → “That’s the billing block we mentioned. Watch — everything else still works.”
- No internet → Open any saved report from Activity; it loads from the device.
- Site slow → Reload once. Pages are statically rendered.
- A judge finds a bug → “Good catch — that’s exactly why we show what we could not verify. We’d rather be corrected than oversell.”